

Critical Coincidence of Alias, Parity, and Gaussian Localization Scales

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July 2026

Abstract

When the counterloop period is tied to the intrinsic endpoint-resolution rank, its first exact alias occurs at the same logarithmic slope as the archived Gaussian localization ceiling. The associated deterministic cycle has clearance $\Theta(\sigma)$, so the old interior Gaussian replacement is already critical. At this order, the isolated alias bias and the scalar square-root parity correction have the same weighted growth exponent $0.4634069445\dots$. We also derive the exact first-alias matching law necessary for a minimal-clock weighted full-trace bridge. Separate absolute majorants remain valid inequalities but do not decay. No divergence of the combined trace error is proved.

1 Natural clock and exact alias

Assume the typed counterloop route chooses

$$k_\sigma = \frac{L_\sigma}{2 \log \lambda} + O(1), \quad L_\sigma = \log(1/\sigma).$$

Its exact moments are

$$s_{k,n} = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n), \quad \beta_k \rightarrow \beta = (0.85\sqrt{\lambda})^{-1}.$$

The limiting pole moment is $p_n = -2\mathbf{1}_{2|n}\beta^n$, and the deterministic numerator anchor is $a_n = c_n - p_n$.

2 Triple critical coincidence

Theorem 2.1. *The first alias order satisfies*

$$2k_\sigma = \frac{L_\sigma}{\log \lambda} + O(1).$$

Its slope equals the archived interior-localization ceiling

$$a_{\text{alias}} = a_{\text{loc}} = \frac{1}{\log \lambda} = 1.9307094191869356\dots$$

The boundary cycle of physical period $2k_\sigma$ has clearance $\Theta(\lambda^{-2k_\sigma}) = \Theta(\sigma)$. Hence its clearance-to-noise ratio has power exponent zero, rather than tending to infinity as required by the existing interior Gaussian proof.

At the same order, both the exact alias bias and the isolated scalar parity correction have weighted growth exponent

$$\chi_1 = \frac{\log(\beta R)}{\log \lambda} = \frac{\log(R/0.85)}{\log \lambda} - \frac{1}{2} = 0.463406944517003\dots$$

Proof. The first display follows directly from the natural clock. RH-10 and RH-17 give a boundary cycle with clearance asymptotic to a positive constant times λ^{-2k} , proving the localization statement.

At $n = 2k$,

$$s_{k,2k} - p_{2k} = (2k - 2)\beta_k^{2k} + 2\beta^{2k}.$$

The finite-radius law

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right]$$

shows that its weighted size divided by $2k$ is $\Theta((\beta R)^{2k})$, with exponent χ_1 .

For the parity eigenvalue, write $\lambda_-(\sigma) = -(1 - \delta_\sigma)$ with $\delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma})$. Since $k_\sigma \delta_\sigma \rightarrow 0$,

$$\frac{|\lambda_-^{2k} - 1|}{2k} \left(\frac{R}{0.85} \right)^{2k} = C_*\sqrt{\sigma} \left(\frac{R}{0.85} \right)^{2k} (1 + o(1)).$$

Its exponent is $\log(R/0.85)/\log \lambda - 1/2 = \chi_1$. □

3 Necessary first-alias matching law

Let

$$e_{\sigma,k,n} = c_{\sigma,n} - s_{k,n} - a_n = (c_{\sigma,n} - c_n) - (s_{k,n} - p_n).$$

The minimal bridge clock lies beyond $2k_\sigma$, so a weighted full-trace estimate includes the first alias.

Corollary 3.1. *If the minimal-clock weighted error tends to zero, then*

$$c_{\sigma,2k} - c_{2k} = (2k - 2)\beta_k^{2k} + 2\beta^{2k} + o(kR^{-2k}).$$

The required absolute matching exponent is

$$\frac{\log R}{\log \lambda} = 0.6496301165394711 \dots$$

Proof. The single alias term of the nonnegative weighted sum must tend to zero, so $e_{\sigma,k,2k} = o(kR^{-2k})$. Substitute the exact alias formula. Since $2k = L_\sigma / \log \lambda + O(1)$, the power of R^{-2k} is the displayed exponent. □

4 Protocol and scope

The computation independently evaluates the alias, parity, matching, and minimal-clock clearance exponents. The first-alias clearance exponent is exactly zero; at the longer minimal bridge clock the clearance ratio decays with exponent $0.4521466691 \dots$

Remark 4.1. Triangle inequalities that majorize alias and parity pieces separately are mathematically valid, but their bounds grow and therefore cannot close the bridge. A joint moving-order boundary-layer trace law could still cancel them. No actual full-trace divergence, spectral-submultiset identification, or Gate A–E conclusion is proved.